Chapter 4: The Structure of Crystalline Solids

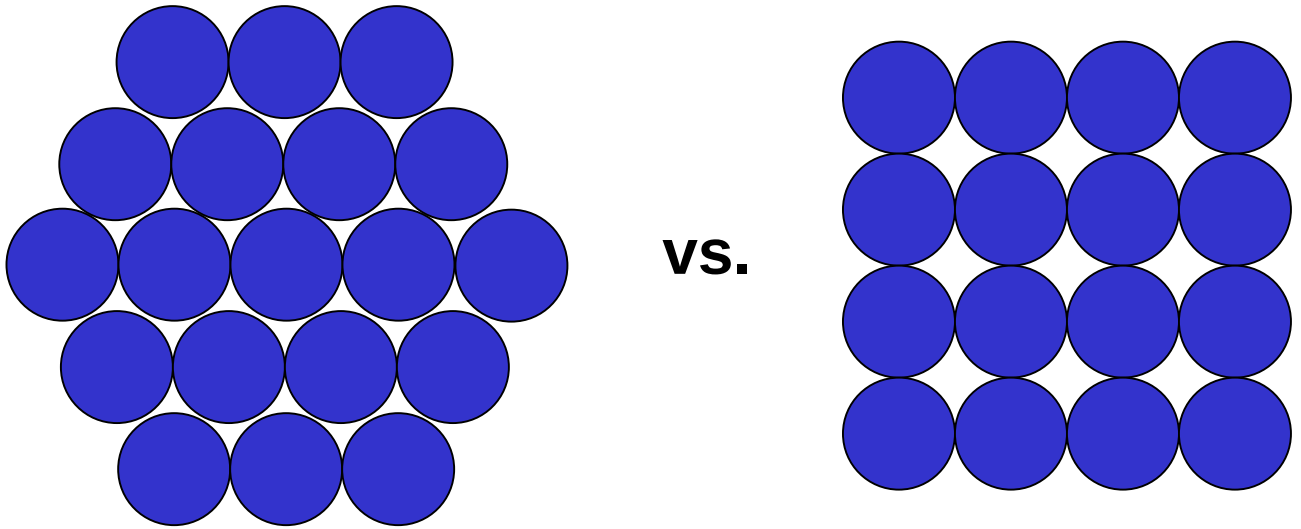
ISSUES TO ADDRESS...

- What are common crystal structures for metals and ceramics?
- What features of a metal's/ceramic's atomic structure determine its density?
- How do the crystal structures of ceramic materials differ from those for metals?

Metallic Crystal Structures

- How can we stack metal atoms to minimize empty space?

2-dimensions



Now stack these 2-D layers to make 3-D structures

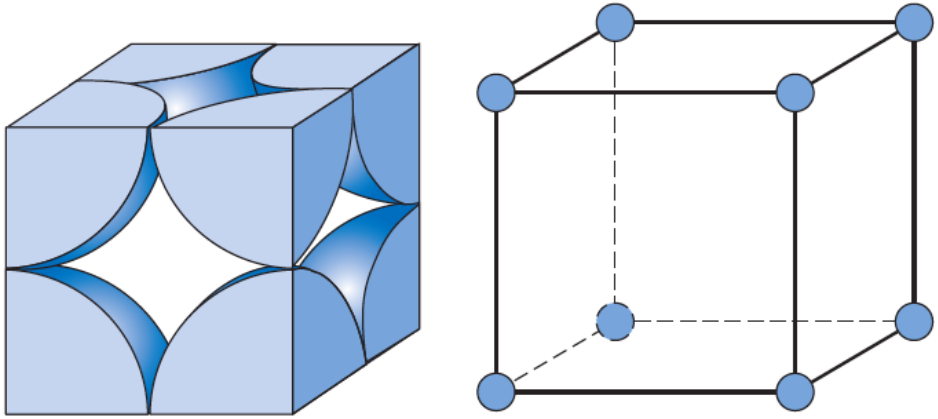
Metallic Crystal Structures

- Tend to be densely packed.
- Reasons for dense packing:
  - Typically, only one element is present, so all atomic radii are the same.
  - Metallic bonding is not directional.
  - Nearest neighbor distances tend to be small in order to lower bond energy.
  - Electron cloud shields cores from each other.
- Metals have the simplest crystal structures.

We will examine three such structures...

Simple Cubic Structure (SC)

- Rare due to low packing density (only Polonium (Po=84) has this structure)
- Close-packed directions are cube edges.
- Coordination # = 6 (# nearest neighbors)



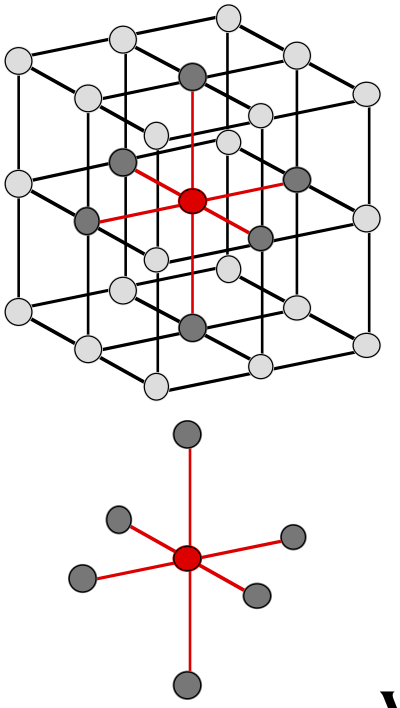


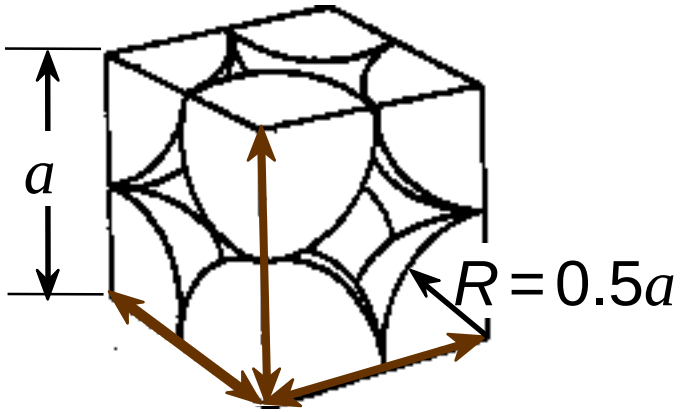
Fig. 4.2, *Callister & Rethwisch 9e.*

Atomic Packing Factor (APF)

$$APF = \frac{\text{Volume of atoms in unit cell*}}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



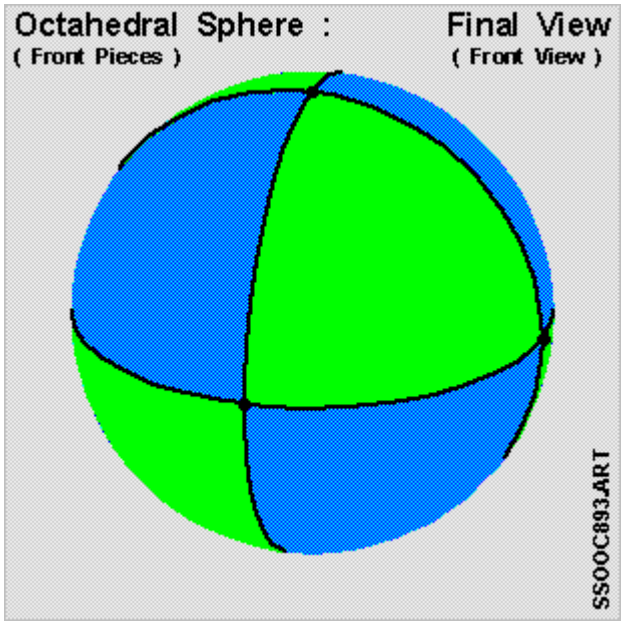
close-packed directions

contains 8 x 1/8 =

1 atom/unit cell

$$APF = \frac{\frac{\text{atoms}}{\text{unit cell}} \cdot \frac{\text{volume}}{\text{atom}}}{\frac{\text{volume}}{\text{unit cell}}} = \frac{1 \cdot \frac{4}{3}\pi (0.5a)^3}{a^3}$$

Adapted from Fig. 4.2 (a), *Callister & Rethwisch 9e.*



$$\text{APF} = \frac{\text{Volume of atoms in unit cell*}}{\text{Volume of unit cell}}$$

Volume of atoms in unit cell =
Number of atoms X Volume of atom

for SC Number of atom = close-packed directions =

contains 8 x 1/8 = 1 atom/unit cell

But a = 2r

Volume of atom = $\frac{4}{3}\pi(0.5a)^3 = \frac{4}{3}\pi r^3$

Volume of Unit cell = $a^3 = (2r)^3 = 8r^3$

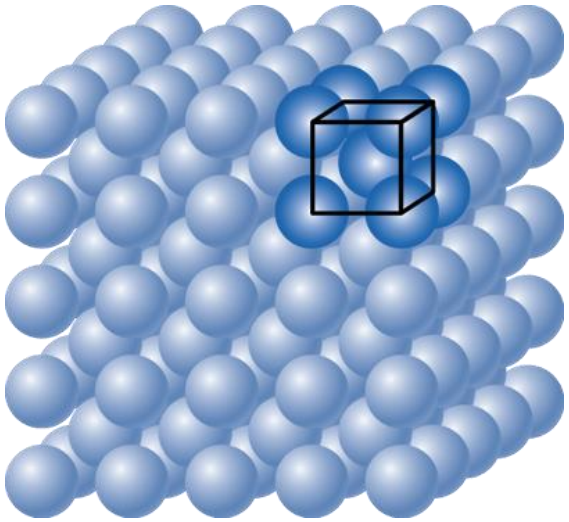
$$APF\ for\ SC = \frac{1 \times \frac{4}{3}\pi r^3}{(2r)^3} = \frac{\frac{4}{3}\pi r^3}{8r^3} = \frac{\frac{4}{3}\pi}{8} = 0.52$$

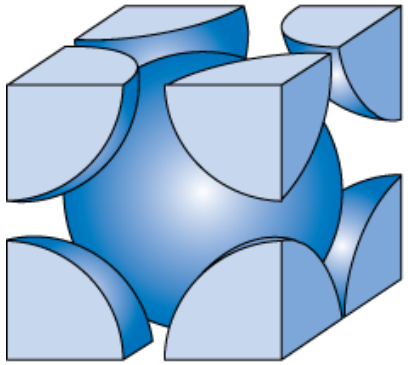
Body Centered Cubic Structure (BCC)

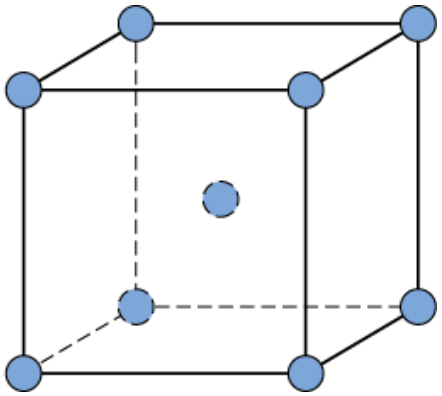
- Atoms touch each other along cube diagonals.
  --Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.

ex: Cr, W, Fe (α), Tantalum, Molybdenum

- Coordination # = 8



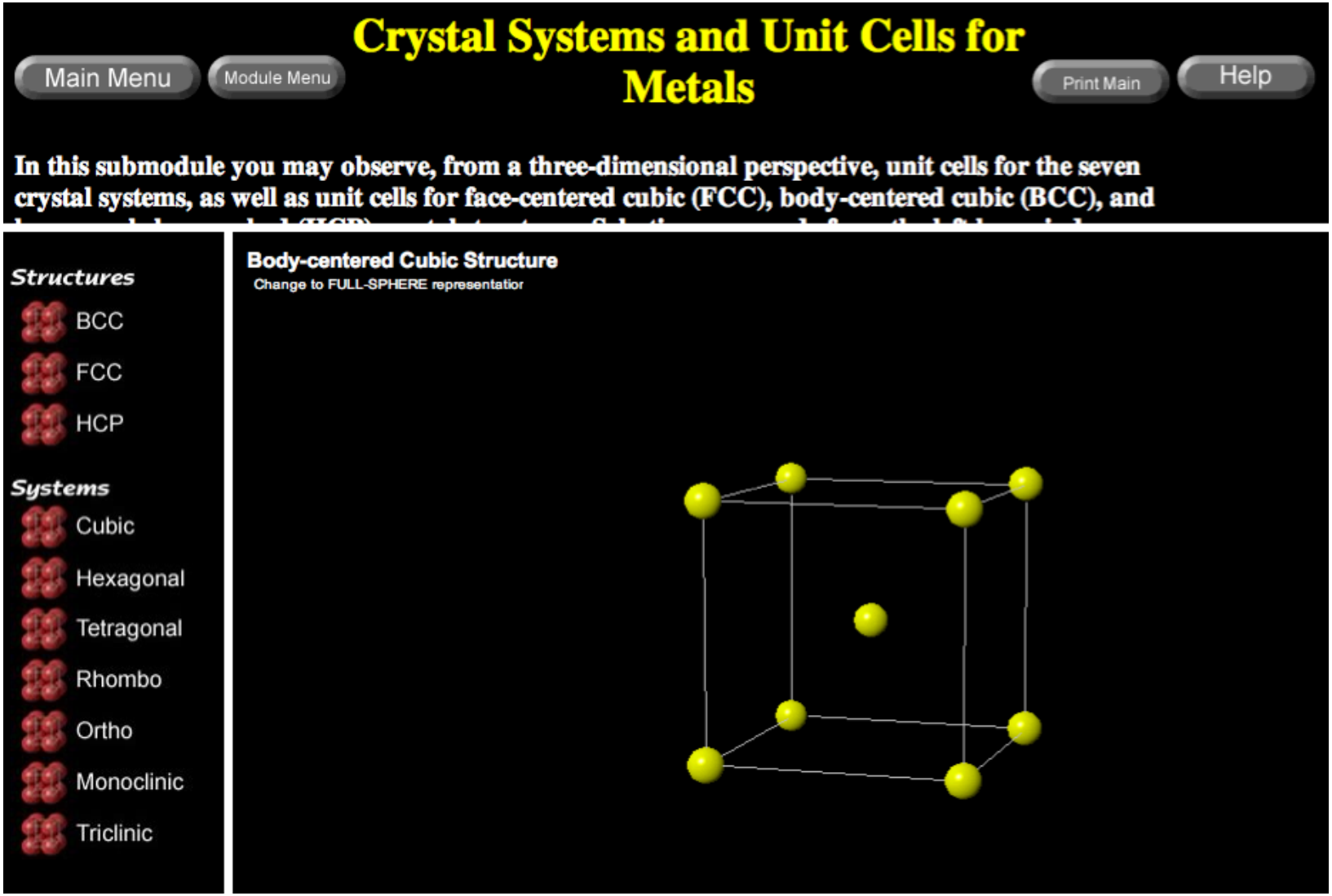




Adapted from Fig. 4.1,
Callister & Rethwisch 9e.

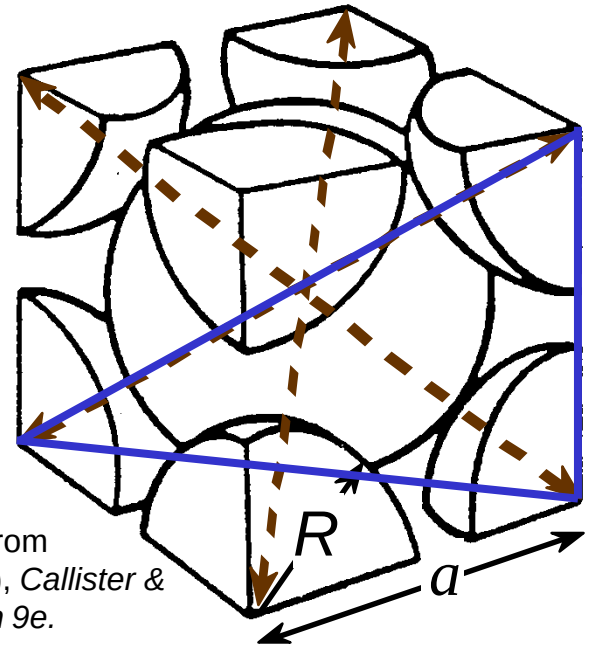
2 atoms/unit cell: 1 center + 8 corners x 1/8

VMSE Screenshot – BCC Unit Cell

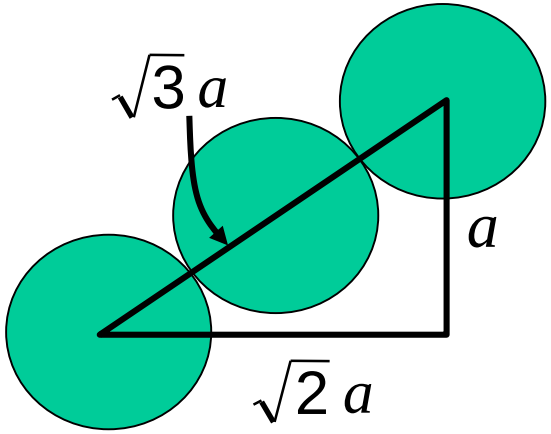


Atomic Packing Factor: BCC

- APF for a body-centered cubic structure = 0.68



Adapted from Fig. 4.1(a), *Callister & Rethwisch 9e.*



Close-packed directions:
length = $4R = \sqrt{3}\,a$

$$APF = \frac{\left(\frac{\text{atoms}}{\text{unit cell}}\right)\left(\frac{\text{volume}}{\text{atom}}\right)}{\left(\frac{\text{volume}}{\text{unit cell}}\right)} = \frac{2\left(\frac{4}{3}\pi\left(\frac{\sqrt{3}a}{4}\right)^3\right)}{a^3}$$

$$APF = \frac{\text{Volume of atoms in unit cell*}}{\text{Volume of unit cell}}$$

Volume of atoms in unit cell = Number of atoms X Volume of atom

for BCC Number of atom = close-packed directions =

Contains 1 atom+ 8 x 1/8 = 2 atom/unit cell

$$but\ 4r = a\sqrt{3}$$

Volume of atom = $\frac{4}{3}\pi(a\frac{\sqrt{3}}{4})^3 = \frac{4}{3}\pi\left(\frac{4r}{\sqrt{3}}\right)^3$

Volume of Unit cell = $a^3 = (\frac{4r}{\sqrt{3}})^3$

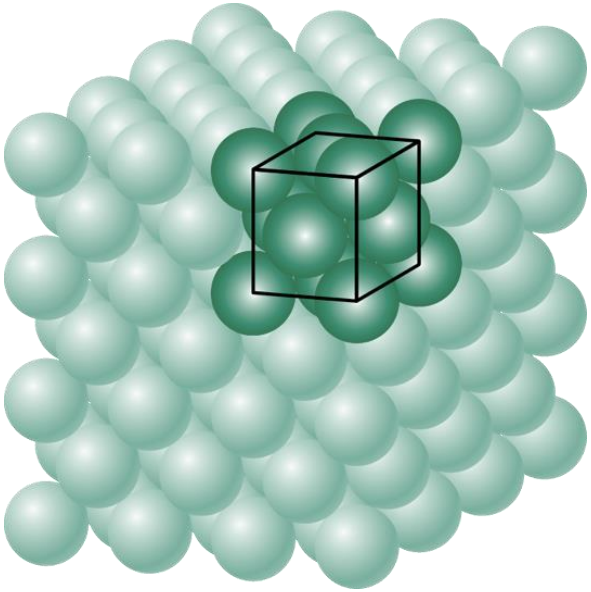
$$APF\ for\ BCC = \frac{2 \times \frac{4}{3}\pi\frac{a^3}{4^3}\sqrt{3}^3}{(a)^3} = 0.68$$

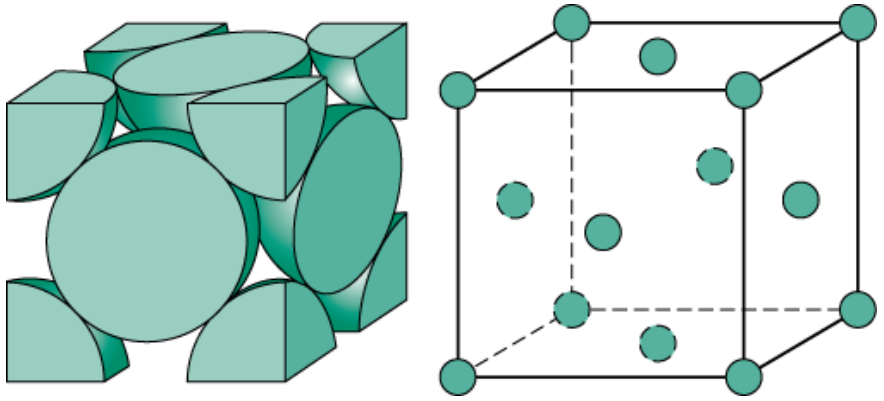
Face Centered Cubic Structure (FCC)

- Atoms touch each other along face diagonals.
  --Note: All atoms are identical; the face-centered atoms are shaded differently only for ease of viewing.

ex: Al, Cu, Au, Pb, Ni, Pt, Ag

- Coordination # = 12



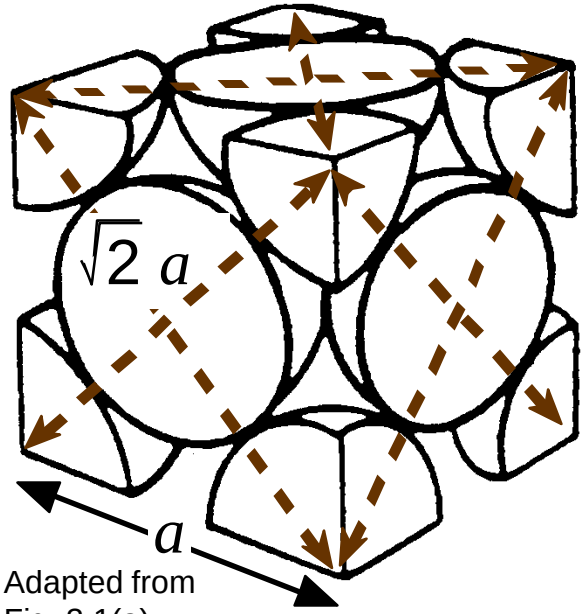


Adapted from Fig. 3.1, *Callister & Rethwisch 9e.*

4 atoms/unit cell: 6 face x 1/2 + 8 corners x 1/8

Atomic Packing Factor: FCC

- APF for a face-centered cubic structure = 0.74
  maximum achievable APF



Adapted from Fig. 3.1(a), *Callister & Rethwisch 9e.*

Close-packed directions:
length = $4R = \sqrt{2}\,a$

Unit cell contains:
6 x 1/2 + 8 x 1/8
= 4 atoms/unit cell

$$\text{APF} = \frac{\dfrac{\text{atoms}}{\text{unit cell}} \cdot \dfrac{\text{volume}}{\text{atom}}}{\dfrac{\text{volume}}{\text{unit cell}}} = \frac{4\left(\frac{4}{3}\pi\left(\frac{\sqrt{2}a}{4}\right)^3\right)}{a^3}$$

$$APF = \frac{\text{Volume of atoms in unit cell*}}{\text{Volume of unit cell}}$$

Volume of atoms in unit cell = Number of atoms X Volume of atom

for FCC Number of atom = close-packed directions =

Contains 6 x 1/2 atom+ 8 x 1/8 = 4 atom/unit cell

$$but\ 4r = a\sqrt{2}$$

Volume of atom = $\frac{4}{3}\pi(a\frac{\sqrt{2}}{4})^3 = \frac{4}{3}\pi\left(\frac{4r}{\sqrt{2}}\right)^3$

Volume of Unit cell = $a^3 = (\frac{4r}{\sqrt{2}})^3$

$$APF\ for\ FCC = \frac{4 \times \frac{4}{3}\pi\frac{a^3}{4^3}\sqrt{2}^3}{(a)^3} = 0.74$$

Theoretical Density, ρ

Is the maximum achievable density of a particular element, compound or alloy, assuming no internal voids or contaminants. It is calculated from the number of atoms per unit cell and measurement of the lattice parameters.

$$\text{Density} = \rho = \frac{\text{Mass of Atoms in Unit Cell}}{\text{Total Volume of Unit Cell}}$$

$$\rho = \frac{n A}{V_C N_A}$$

where n = number of atoms/unit cell

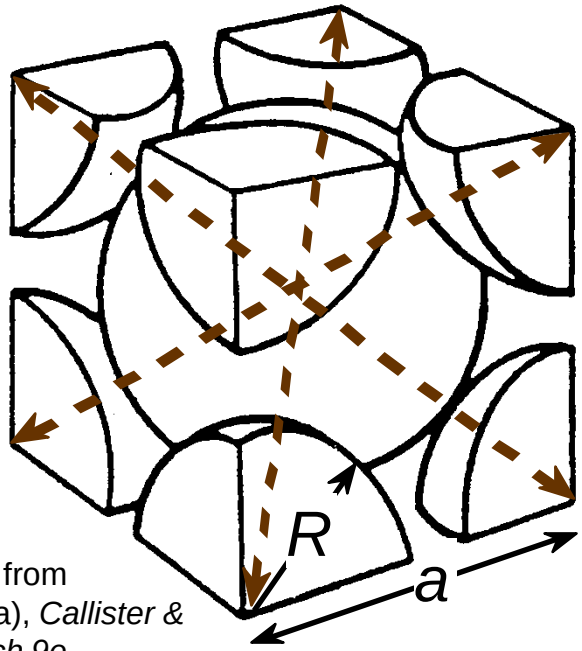
A = atomic weight

V_C = Volume of unit cell = a^3 for cubic

N_A = Avogadro's number

= 6.022×10^{23} atoms/mol

Theoretical Density, ρ



Adapted from Fig. 4.1(a), *Callister & Rethwisch 9e.*

- Ex: Cr (BCC)

  A = 52.00 g/mol

  R = 0.125 nm

  n = 2 atoms/unit cell

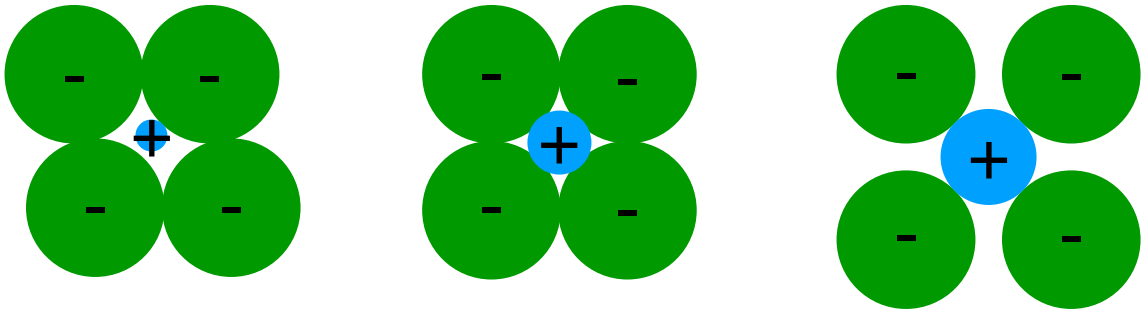
$a = 4R/\sqrt{3} = 0.2887$ nm

$$\rho = \frac{\dfrac{\text{atoms}}{\text{unit cell}} \cdot \dfrac{\text{g}}{\text{mol}}}{\text{volume/unit cell} \cdot \dfrac{\text{atoms}}{\text{mol}}} = \frac{2 \cdot 52.00}{a^3 \cdot 6.022 \times 10^{23}}$$

$\rho_{\text{theoretical}}$	= 7.18 g/cm^3
ρ_{actual}	= 7.19 g/cm^3

Factors that Determine Crystal Structure

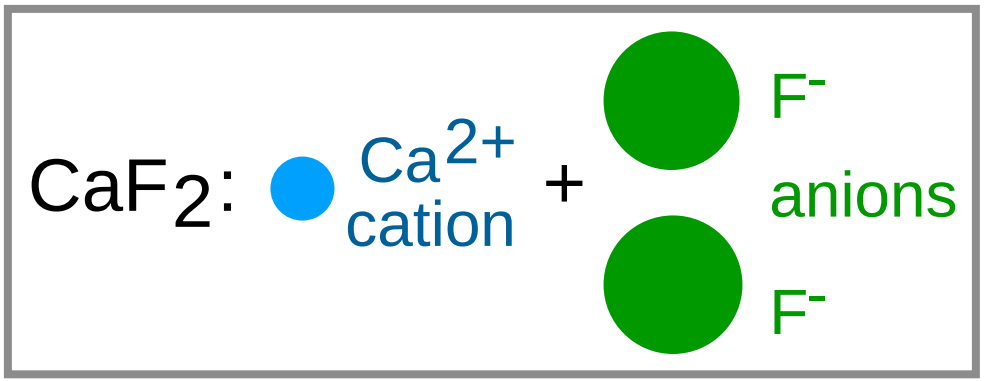
1. Relative sizes of ions – Formation of stable structures:
   --maximize the # of oppositely charged ion neighbors.



unstable stable stable

Adapted from Fig. 4.4, *Callister & Rethwisch 9e.*

2. Maintenance of Charge Neutrality :
   --Net charge in ceramic should be zero.
   --Reflected in chemical formula:



A_mX_p

m, p values to achieve charge neutrality

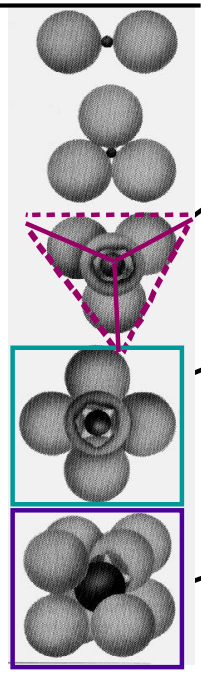
Coordination Number and Ionic Radii

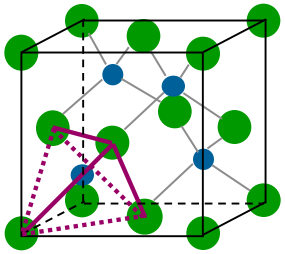
- Coordination Number increases with $\frac{r_{cation}}{r_{anion}}$

To form a stable structure, how many anions can surround around a cation?

$\frac{r_{cation}}{r_{anion}}$	Coord. Number	
< 0.155	2	linear
0.155 - 0.225	3	triangular
0.225 - 0.414	4	tetrahedral
0.414 - 0.732	6	octahedral
0.732 - 1.0	8	cubic

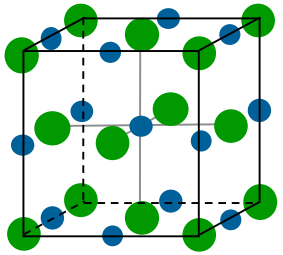
Adapted from Table 4.3, *Callister & Rethwisch 9e.*





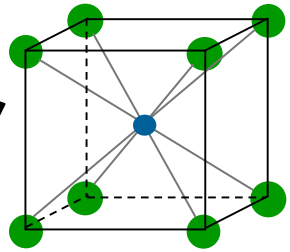
ZnS (zinc blende)

Adapted from Fig. 4.7, *Callister & Rethwisch 9e.*



NaCl (sodium chloride)

Adapted from Fig. 4.5, *Callister & Rethwisch 9e.*



CsCl (cesium chloride)

Adapted from Fig. 4.6, *Callister & Rethwisch 9e.*

Example Problem: Predicting the Crystal Structure of FeO

- On the basis of ionic radii, what crystal structure would you predict for FeO?

Cation	Ionic radius (nm)
Al^{3+}	0.053
Fe^{2+}	0.077
Fe^{3+}	0.069
Ca^{2+}	0.100

Anion	
O^{2-}	0.140
Cl^{-}	0.181
F^{-}	0.133

- Answer:

$$\frac{r_{cation}}{r_{anion}} = \frac{0.077}{0.140} = 0.550$$

based on this ratio,
-- coord # = 6 because

$0.414 < 0.550 < 0.732$

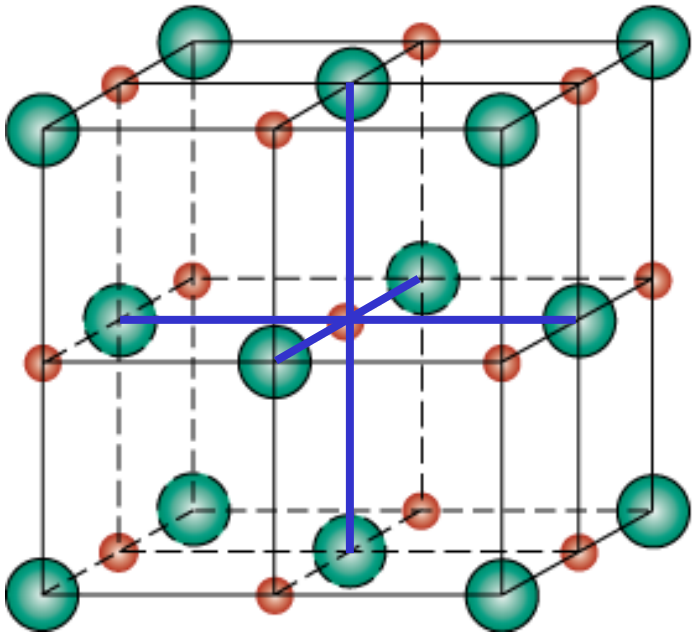
-- crystal structure is Octahedral

Data from Table 4.4,
Callister & Rethwisch 9e.

Rock Salt Structure

Same concepts can be applied to ionic solids in general.

Example: NaCl (rock salt) structure



Adapted from Fig. 4.5,
Callister & Rethwisch 9e.

Na^+ $r_{Na} = 0.102$ nm

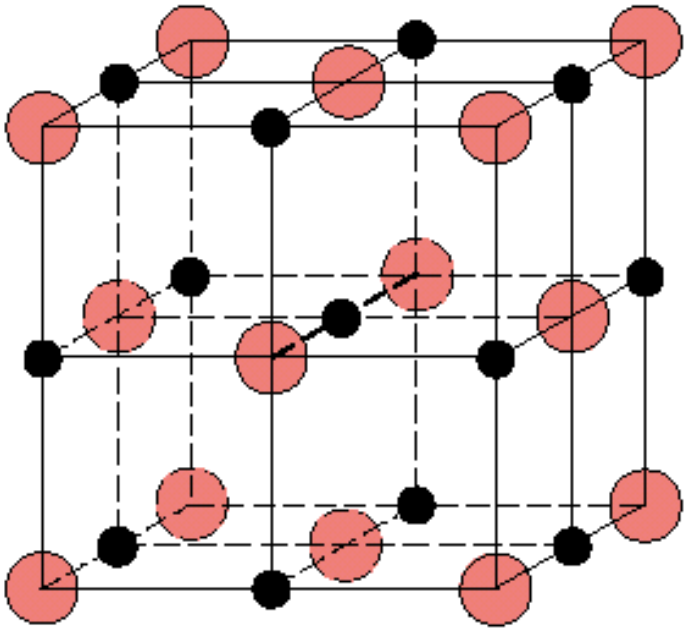
Cl^- $r_{Cl} = 0.181$ nm

$r_{Na}/r_{Cl} = 0.564$

$\therefore$ cations (Na^+) prefer octahedral sites

MgO and FeO

MgO and FeO also have the NaCl structure



Adapted from Fig. 4.5,
Callister & Rethwisch 9e.

O^{2-} $r_O = 0.140$ nm

Mg^{2+} $r_{Mg} = 0.072$ nm

$r_{Mg}/r_O = 0.514$

$\therefore$ cations prefer octahedral sites

So each Mg^{2+} (or Fe^{2+}) has 6 neighbor oxygen atoms

AX Crystal Structures

AX–Type Crystal Structures include NaCl, CsCl, and zinc blende

Cesium Chloride structure:

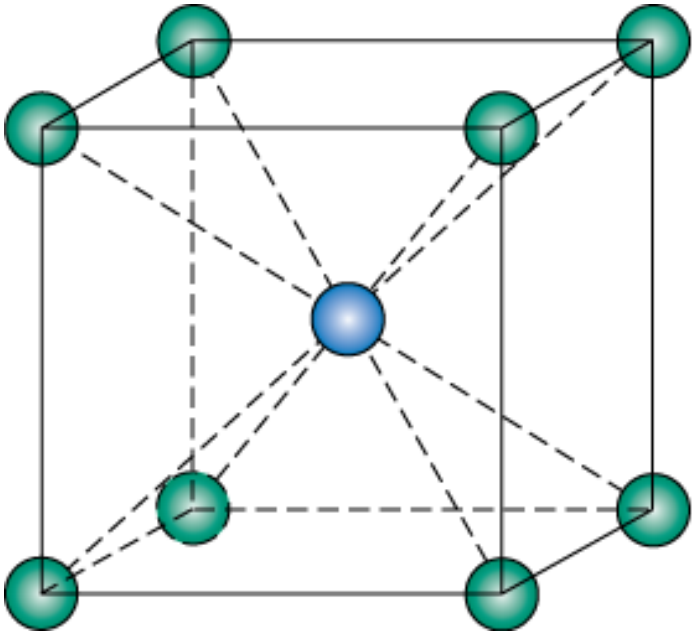


Fig. 4.6, *Callister & Rethwisch 9e.*

$$\frac{r_{Cs^+}}{r_{Cl^-}} = \frac{0.170}{0.181} = 0.939$$

$\therefore$ Since 0.732 < 0.939 < 1.0, cubic sites preferred

So each Cs^+ has 8 neighbor Cl^-

AX_2 Crystal Structures

Fluorite structure

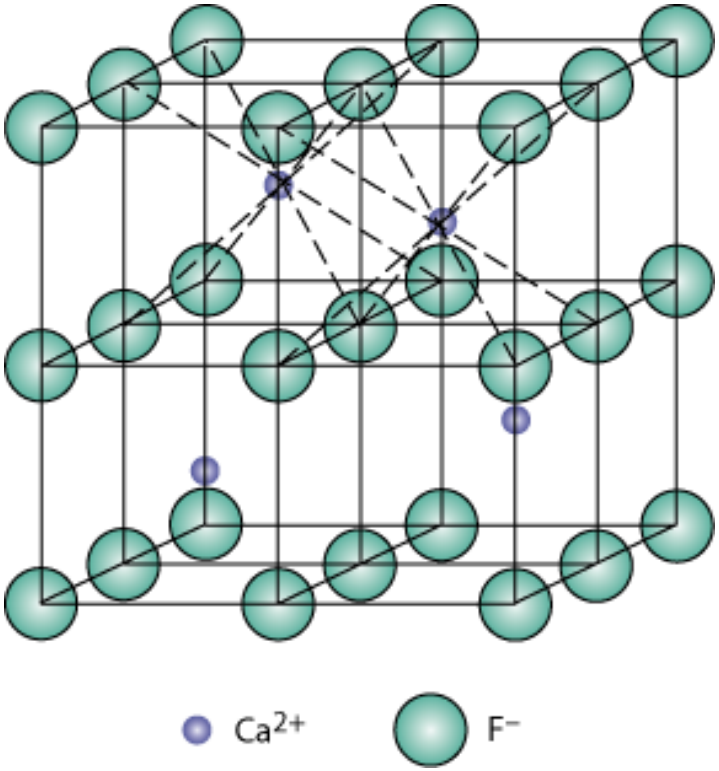


Fig. 4.8, *Callister & Rethwisch 9e.*

- Calcium Fluorite (CaF_2)
- Cations in cubic sites

- UO_2, ThO_2, ZrO_2, CeO_2

- Antifluorite structure – positions of cations and anions reversed

ABX_3 Crystal Structures

- Perovskite structure

Ex: complex oxide

$BaTiO_3$

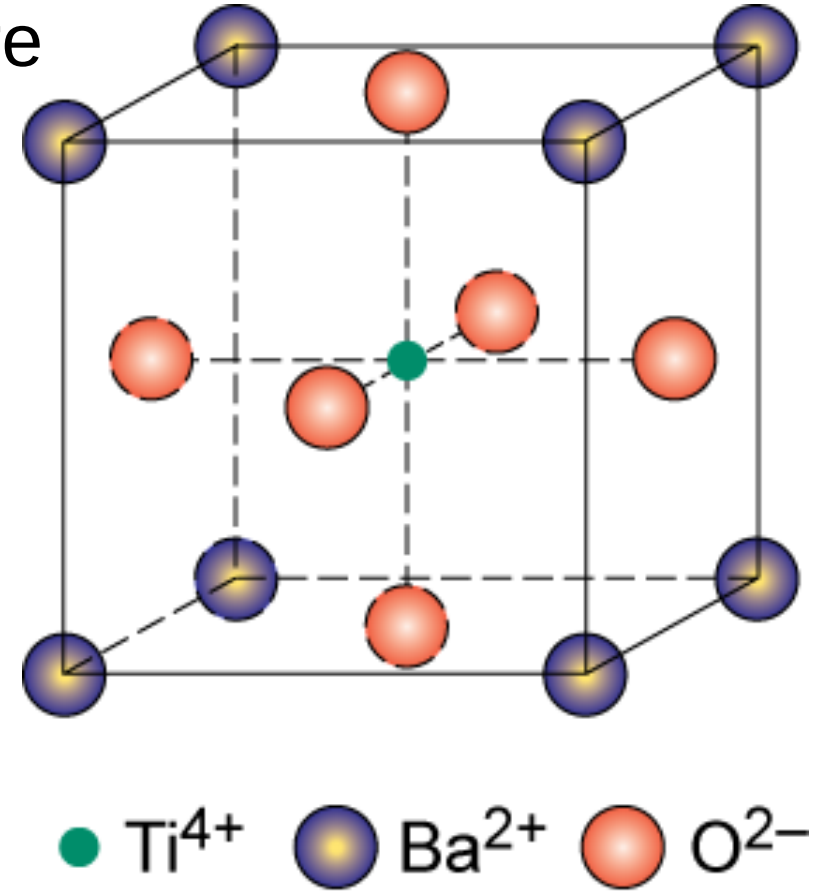


Fig. 4.9, *Callister & Rethwisch 9e.*

Densities of Material Classes

In general

$\rho_{metals} > \rho_{ceramics} > \rho_{polymers}$

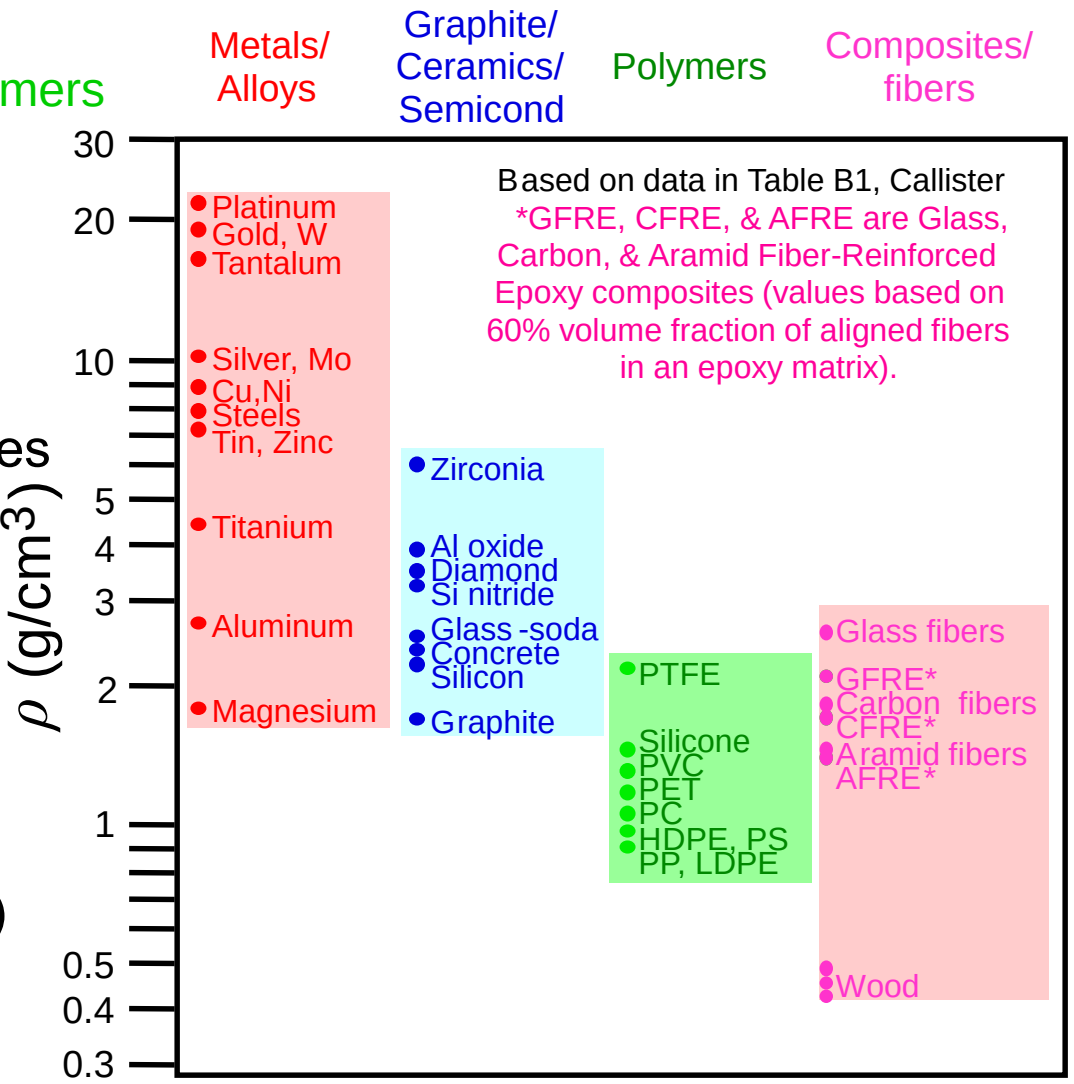
Why?

Metals have...
- close-packing (metallic bonding)
- often large atomic masses

Ceramics have...
- less dense packing
- often lighter elements

Polymers have...
- low packing density (often amorphous)
- lighter elements (C,H,O)

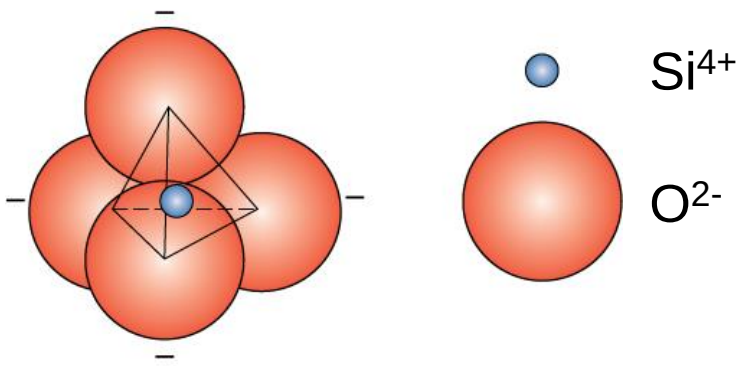
Composites have...
- intermediate values



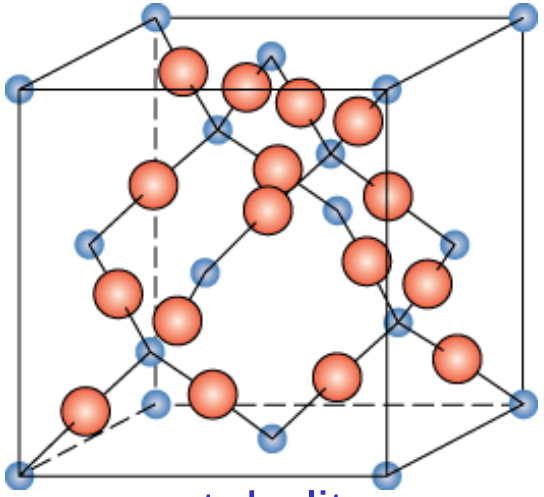
Data from Table B.1, *Callister & Rethwisch, 9e.*

Silicate Ceramics

Most common elements on earth are Si & O



Figs. 4.10 & 4.11, *Callister & Rethwisch 9e*

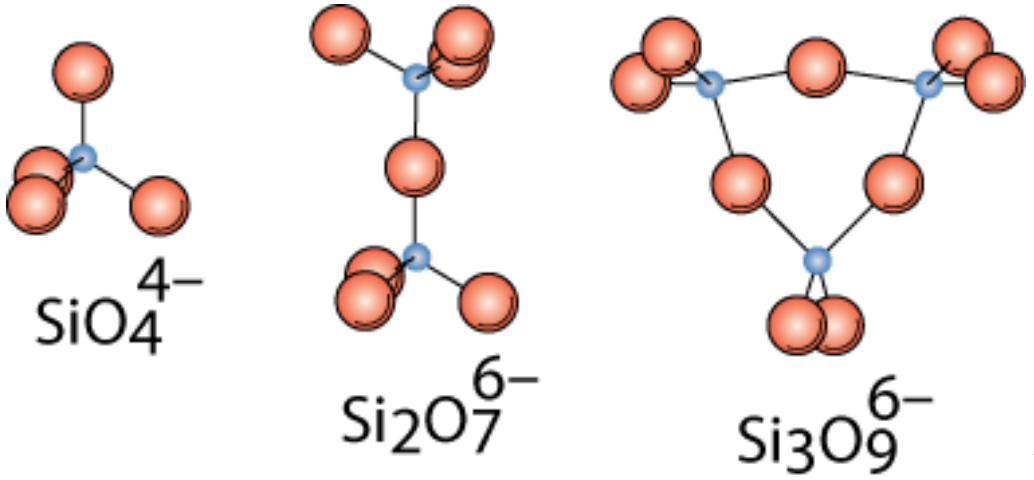


crystobalite

- SiO_2 (silica) polymorphic forms are quartz, crystobalite, & tridymite
- The strong Si-O bonds lead to a high melting temperature (1710ºC) for this material

Silicates

Bonding of adjacent SiO_4^{4-} accomplished by the sharing of common corners, edges, or faces



Adapted from Fig. 4.13, *Callister & Rethwisch 9e.*

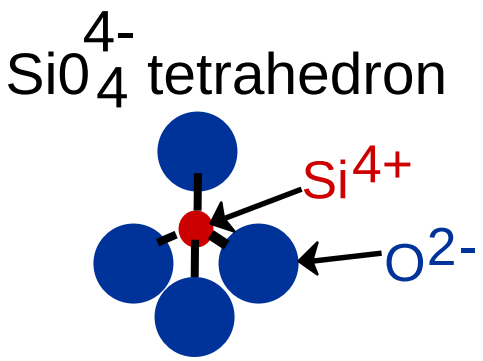
Mg_2SiO_4 $Ca_2MgSi_2O_7$

Presence of cations such as Ca^{2+}, Mg^{2+}, & Al^{3+}

1. maintain charge neutrality, and
2. ionically bond SiO_4^{4-} to one another

Glass Structure

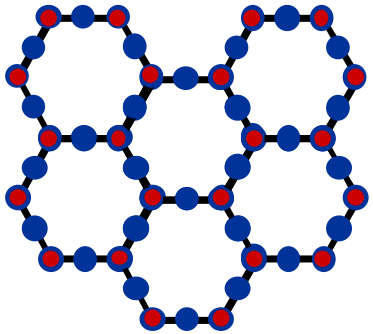
- Basic Unit:

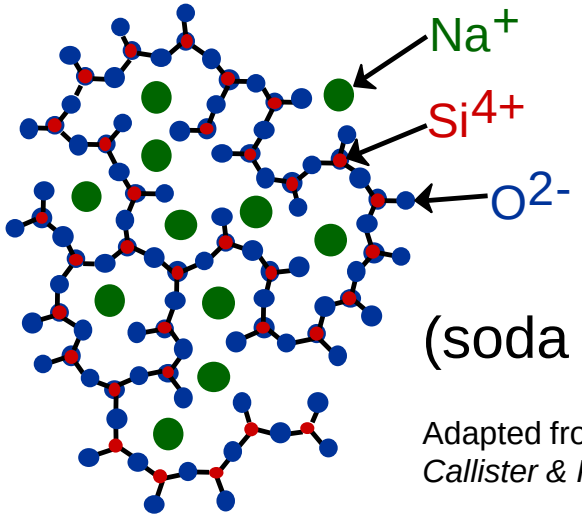


Glass is noncrystalline (amorphous)

- Fused silica is SiO_2 to which no impurities have been added
- Other common glasses contain impurity ions such as Na^+, Ca^{2+}, Al^{3+}, and B^{3+}

- Quartz is crystalline SiO_2:





(soda glass)

Adapted from Fig. 4.12, *Callister & Rethwisch 9e*.

Polycrystals

- *Most* engineering materials are polycrystals.



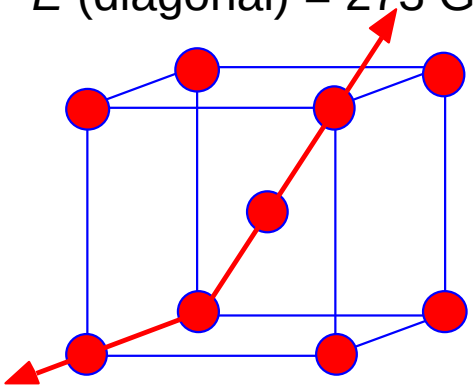
Fig. K, color inset pages of *Callister 5e*.
(Courtesy of Paul E. Danielson, Teledyne Wah Chang Albany)

- Nb-Hf-W plate with an electron beam weld.
- Each "grain" is a single crystal.
- If grains are randomly oriented,
  overall component properties are not directional.
- Grain sizes typically range from 1 nm to 2 cm
  (i.e., from a few to millions of atomic layers).

Single vs Polycrystals

- Single Crystals
  - -Properties vary with direction: anisotropic.
  - -Example: the modulus of elasticity (E) in BCC iron:

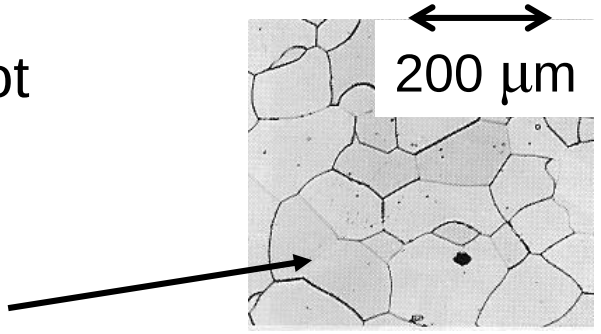
E (diagonal) = 273 GPa

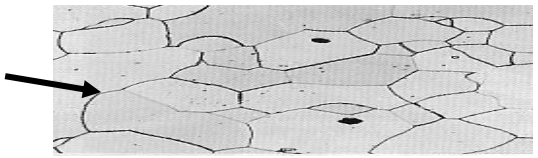


E (edge) = 125 GPa

Data from Table 3.3, *Callister & Rethwisch 9e*. (Source of data is R.W. Hertzberg, *Deformation and Fracture Mechanics of Engineering Materials*, 3rd ed., John Wiley and Sons, 1989.)

- Polycrystals
  - -Properties may/may not vary with direction.
  - -If grains are randomly oriented: isotropic. ($E_{\text{poly iron}}$ = 210 GPa)
  - -If grains are textured, anisotropic.





Adapted from Fig. 6.19(b), *Callister & Rethwisch 9e*. [Fig. 6.19(b) is courtesy of L.C. Smith and C. Brady, the National Bureau of Standards, Washington, DC (now the National Institute of Standards and Technology, Gaithersburg, MD).]

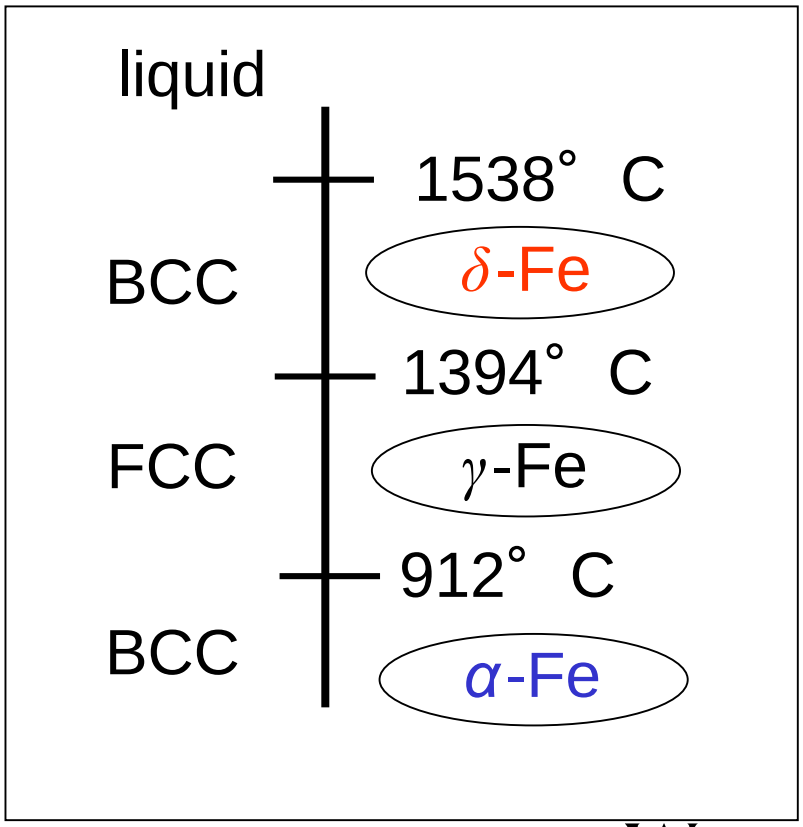
Polymorphism

polymorphism, in crystallography, the condition in which a material exists in more than one crystalline form; the forms differ somewhat in physical and, sometimes, chemical properties.

Two or more distinct crystal structures for the same material (allotropy/polymorphism)

Titanium exists as *α-Ti and β* -Ti

Carbon exists as diamond and graphite

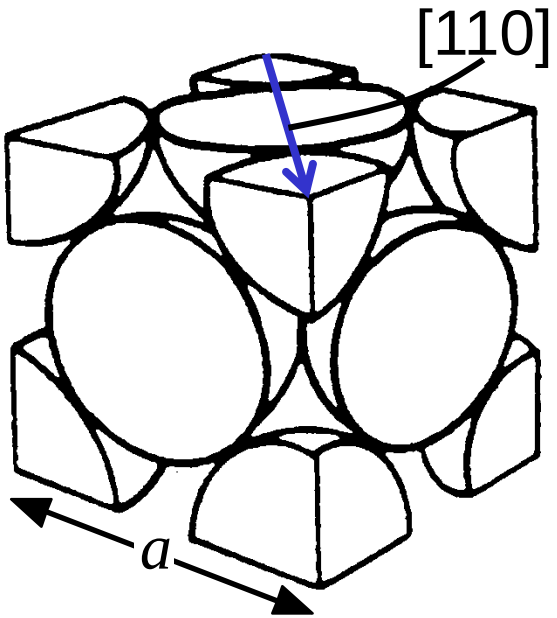


Linear Density

The fraction of line length intersected by atoms is equal to the linear density while Crystallographic planes that are equivalent have the same atomic planar density. The plane of interest is positioned so as to pass through atom centers

Linear Density

- Linear Density of Atoms ≡ LD = $\dfrac{\text{Number of atoms}}{\text{Unit length of direction vector}}$



Adapted from Fig. 3.1(a), *Callister & Rethwisch 9e.*

ex: linear density of Al in [110] direction

a = 0.405 nm

atoms → 2

length → $\sqrt{2}a$

$$LD = \frac{2}{\sqrt{2}a} = 3.5\ \text{nm}^{-1}$$

Planar Density of (100) Iron

Solution: At T < 912˚ C iron has the BCC structure.

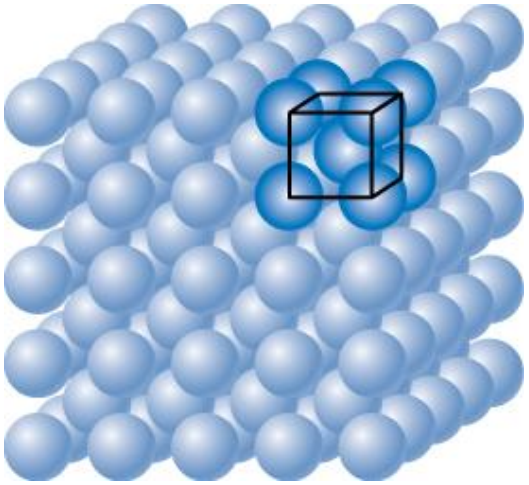
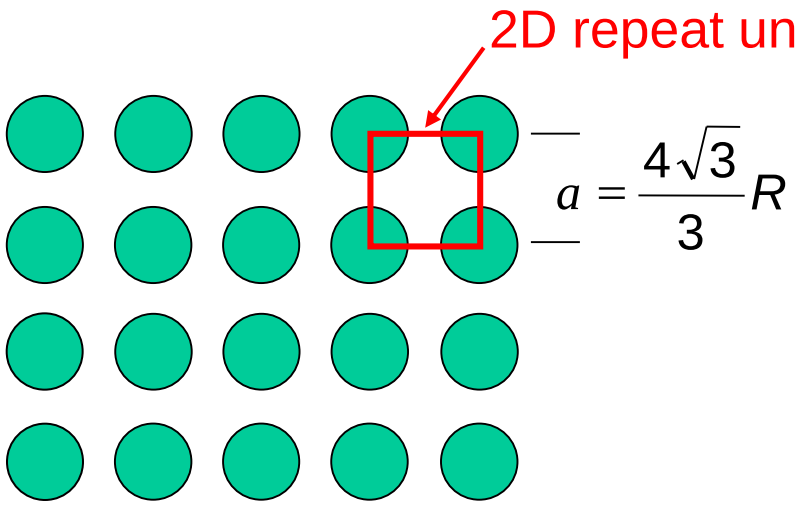


Fig. 4.2(c), *Callister & Rethwisch 9e* [from W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, p. 51. Copyright © 1964 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.]

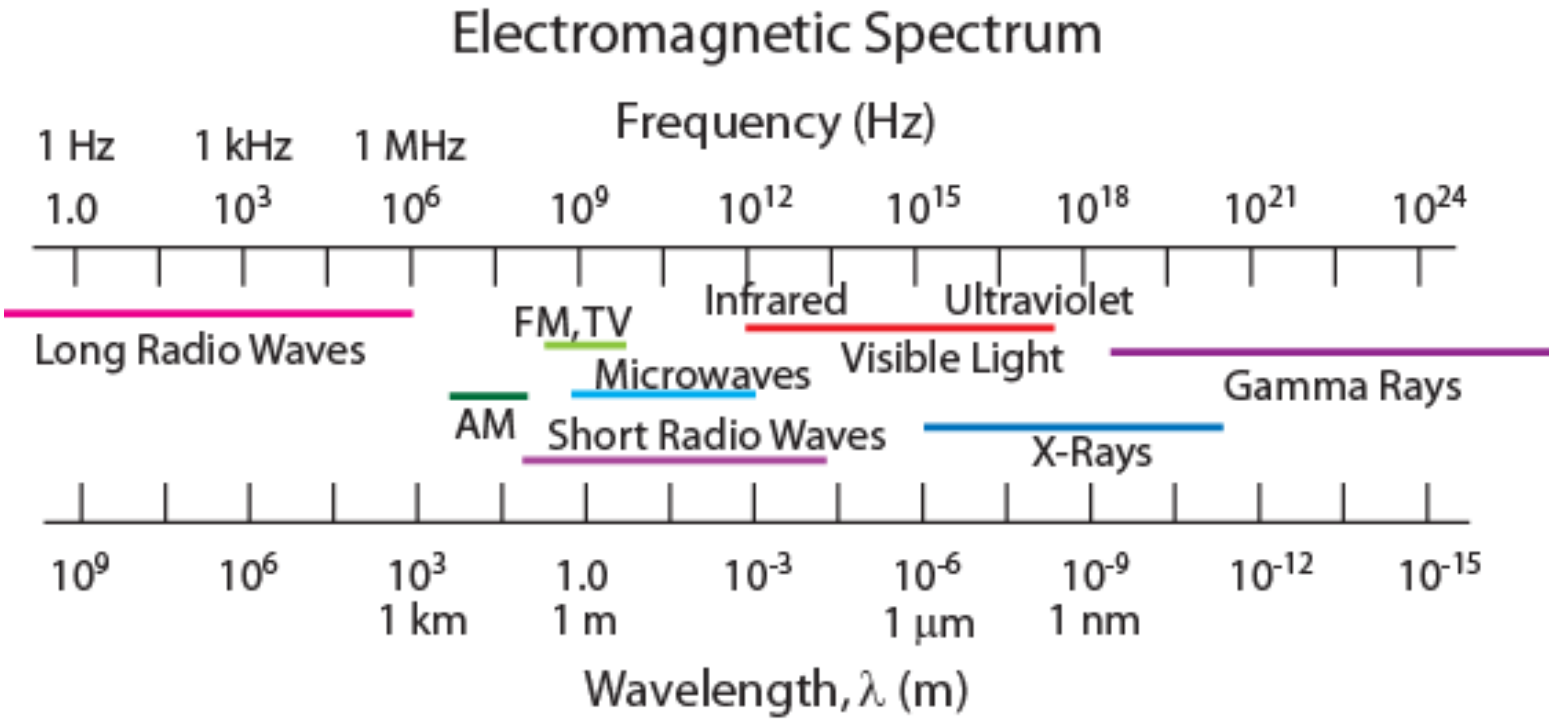
(100)



Radius of iron R = 0.1241 nm

$$\text{Planar Density} = \frac{\dfrac{\text{atoms}}{\text{2D repeat unit}}}{\dfrac{\text{area}}{\text{2D repeat unit}}} = \frac{1}{a^2} = \frac{1}{\left(\frac{4\sqrt{3}}{3}R\right)^2} = 12.1\ \frac{\text{atoms}}{\text{nm}^2} = 1.2 \times 10^{19}\ \frac{\text{atoms}}{\text{m}^2}$$

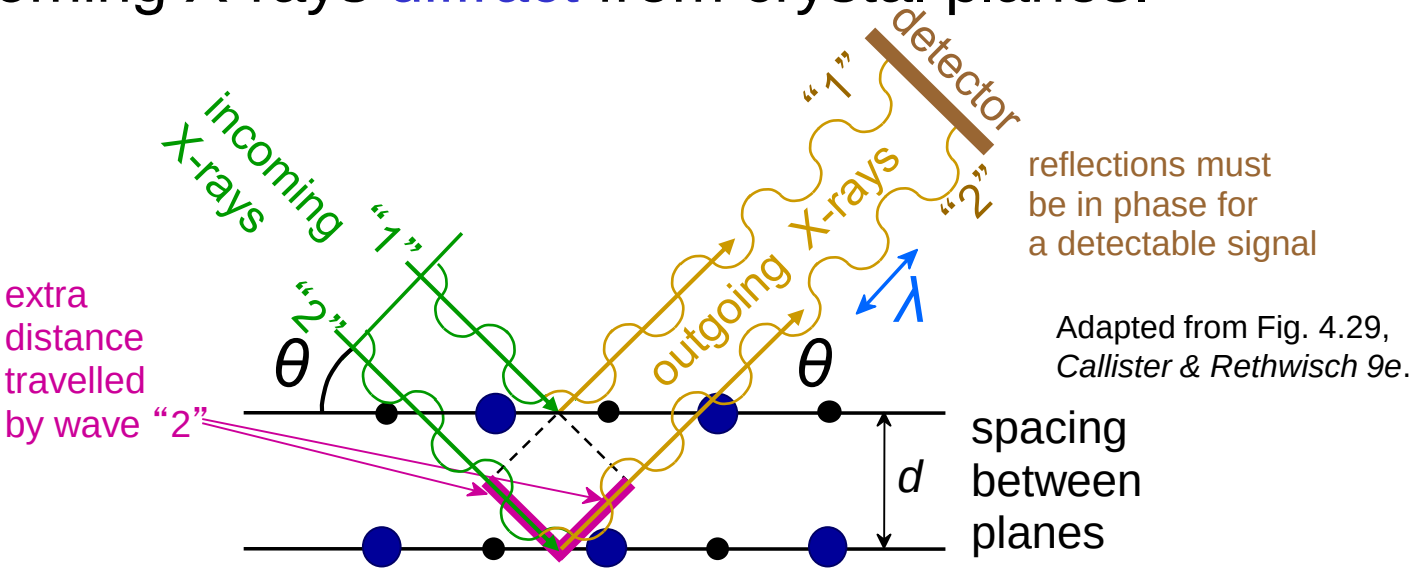
X-Ray Diffraction



- Diffraction gratings must have spacings comparable to the wavelength of diffracted radiation.
- Can't resolve spacings $< \lambda$
- Spacing is the distance between parallel planes of atoms.

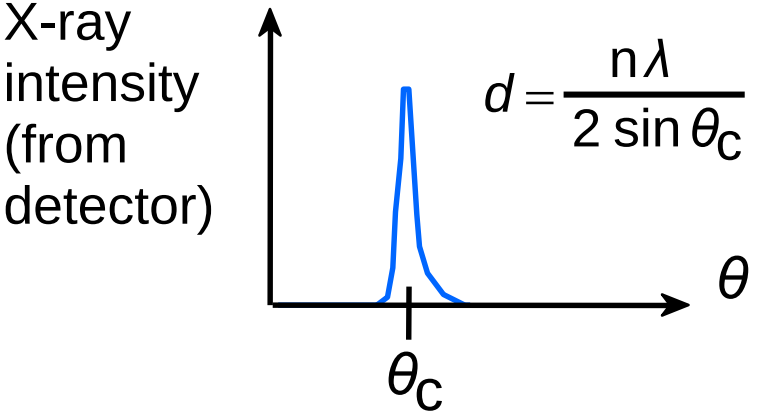
X-Rays to Determine Crystal Structure

- Incoming X-rays diffract from crystal planes.

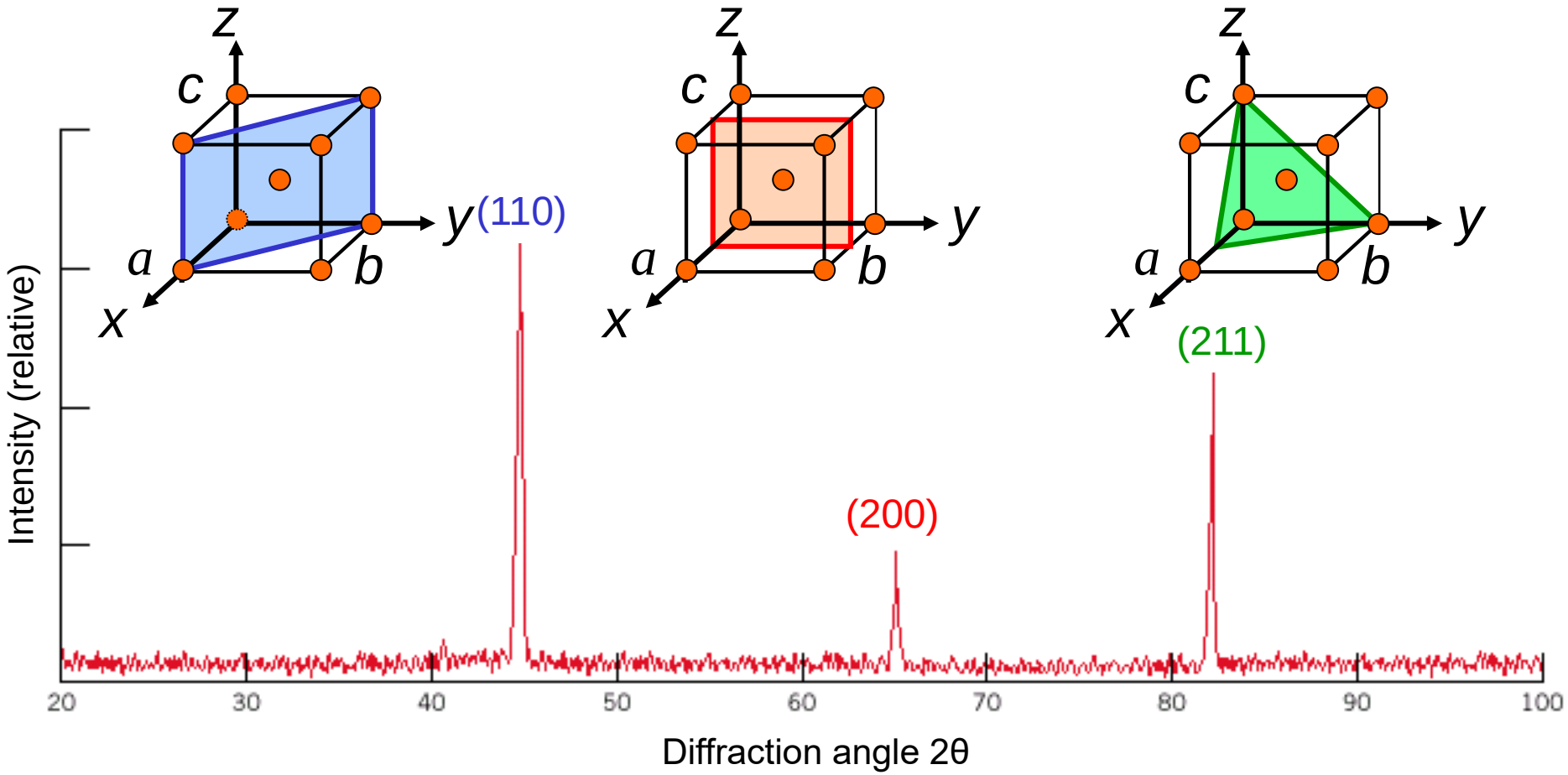


Adapted from Fig. 4.29, *Callister & Rethwisch 9e*.

Measurement of critical angle, θ_c, allows computation of planar spacing, d.



X-Ray Diffraction Pattern



Diffraction pattern for polycrystalline α-iron (BCC)

Adapted from Fig. 3.22, *Callister 8e.*

Summary

- Common metallic crystal structures are FCC, BCC, and HCP. Coordination number and atomic packing factor are the same for both FCC and HCP crystal structures.
- We can predict the density of a material, provided we know the atomic weight, atomic radius, and crystal geometry (e.g., FCC, BCC, HCP).
- Interatomic bonding in ceramics is ionic and/or covalent.
- Ceramic crystal structures are based on:
  -- maintaining charge neutrality
  -- cation-anion radii ratios.
- Some materials can have more than one crystal structure. This is referred to as polymorphism (or allotropy).
- X-ray diffraction is used for crystal structure and interplanar spacing determinations.

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: